

Table 5: Spectral entropy, L0 sparsity, and absolute magnitudes
AdamW has lower spectral entropy (more concentrated updates) than Muon across all conditions

Optimizer	Obj.	Prec.	Spectral Entropy	$\ \Delta W\ _F$	$\Sigma \sigma_i$	Rank @90%	% SVs < 1e-5	% SVs < 1e-4	σ_1	mean(σ_i)
AdamW	GRPO	FP32	0.606	0.0327	0.6399	190	1.8%	14.7%	1.44e-02	5.04e-04
AdamW	GRPO	BF16	0.947	0.0063	0.2156	732	2.4%	32.8%	4.87e-04	1.65e-04
AdamW	GRPO	Mixed	0.724	0.0334	0.8069	329	1.3%	11.1%	1.23e-02	6.31e-04
Muon	GRPO	FP32	0.962	0.0214	0.7469	752	1.3%	7.8%	1.12e-03	5.78e-04
Muon	GRPO	BF16	0.888	0.0014	0.0435	590	19.2%	96.6%	1.95e-04	3.63e-05
Muon	GRPO	Mixed	0.960	0.0209	0.7271	748	1.3%	7.8%	1.19e-03	5.66e-04
AdamW	SFT	FP32	0.789	0.0387	1.0558	450	1.1%	8.5%	1.16e-02	8.16e-04
AdamW	SFT	BF16	0.942	0.0187	0.6367	717	0.9%	8.5%	1.67e-03	4.90e-04
AdamW	SFT	Mixed	0.720	0.0467	1.1533	379	0.9%	7.6%	1.67e-02	8.96e-04
Muon	SFT	FP32	0.967	0.0333	1.1756	794	0.8%	4.2%	1.81e-03	9.09e-04
Muon	SFT	BF16	0.839	0.0019	0.0543	492	19.7%	89.7%	3.73e-04	4.75e-05
Muon	SFT	Mixed	0.950	0.0321	1.0981	730	1.0%	6.0%	2.05e-03	8.66e-04